

ICT Classified -Chapter (7) - System life cycle

1

The owner of a sports club has recently had a new computerised stock control system installed. He employed a systems analyst to research the existing system and then install the new system.

- (a) Tick the relevant stage of the systems analysis and design (systems life cycle) for each of the following activities.

Activity	Analysis ✓	Design ✓	Evaluation ✓
Interviewing the users of the new system			
Interviewing the users of the existing system			
Planning the validation routines			
Examining existing documents			

[4]

- (b) Before the system was installed, the systems analyst had to decide on a method of implementation.

Name and describe **three** methods of implementation.

Name

Description

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Name

Description.....

.....

Name.....

Description

.....

[6]

- (c) When the system was implemented, the system analyst gave some documentation to the sports club owner.

Name **three** items found in the technical documentation of this system.

1.
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2.
.....
3.
.....

[3]

- (d) Name **three** items in the user documentation which are not present in the technical documentation.

1.
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2.
.....
3.
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[3]

Q 8 - Page 6 0417/11 May/June 2015

2

Below is a list of statements. Each statement is true for only one of the given methods of implementation. Tick the method which matches the statement.

	Parallel running ✓	Direct Changeover ✓	Pilot running ✓
All of the old and new systems run at the same time.			
If the system fails in one branch, the rest of the company is not affected.			
The new system has to be completely free of errors before implementation.			
The benefits of the new system are available immediately.			

[4]

Q2 - Page 3 0417/12 May/June 2015

3

Tick whether the following items are found in user documentation only, technical documentation only or are found in both.

	User ✓	Technical ✓	Both ✓
Systems flowchart			
How to save a document			
List of variables			
The purpose of the system			

[4]

Q4 - Page 3 0417/12 May/June 2015

Describe the different testing strategies that can be used on a new system.

[6]

Q15 – Page 12 0417/12 May/June 2015

	Verification ✓	Validation ✓	Neither verification nor validation ✓
Data is entered by two different operators			
Data is checked to see if it is present			
Data is checked to make sure it is correct			
Data entered is checked to see if it matches data on the source document			

[4]

Q4 - Page 3 0417/13 May/June 2015

6

An airline keeps a database of all the flights it operates. A short extract is shown below.

Flight_number	Departure_Airport_Code	Number_of_passengers	Destination_Airport_code	Ticket_prices
EK236	JNB	135	LHR	\$1200
EK080	DXB	256	DEL	\$500
EK029	LHR	375	DXB	\$650

(a) Explain fully, what is meant by the following validation checks using the extract above.

Range check

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Format check

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Length check

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.....

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[6]

(b) Using the extract above, give an example of **one** field which would contain:

Text data
(alphanumeric)

Numeric data

[2]

Q12 - Page 10 0417/13 May/June 2015

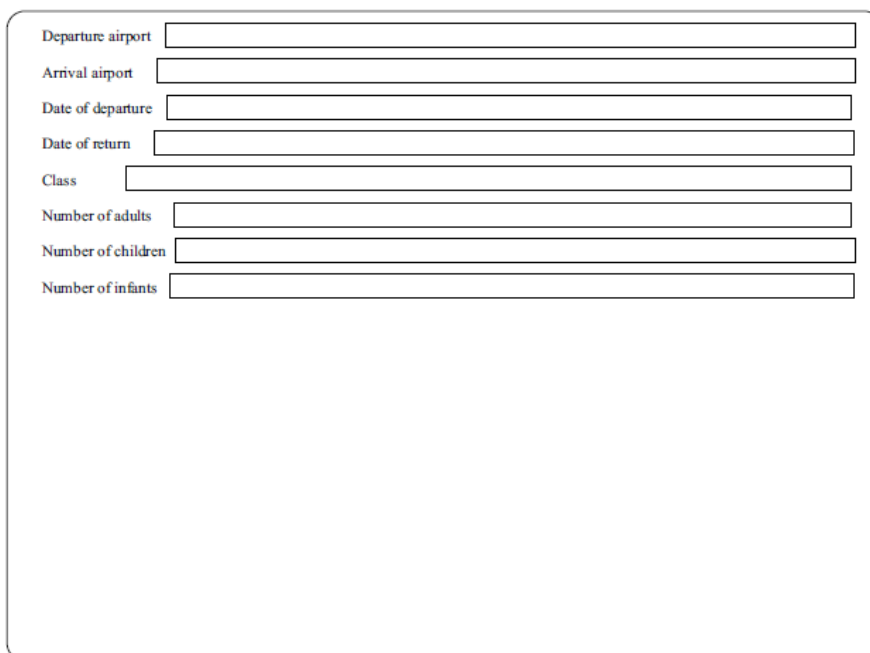
7

An airline company allows customers to book flights online.

Here is an example of the data a customer is asked to type in when they search for a suitable flight, together with a typical entry. There are three types of class: Economy, Business and First.

Departure airport	LHR
Arrival airport	DXB
Date of departure	21/9/2016
Date of return	8/10/2016
Class	Economy
Number of adults	2
Number of children	2
Number of infants	1

Below is an input screen designed to help this data to be entered.



(a) Evaluate this form regarding the ease of entering data

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[6]

(b) Explain why it would not be appropriate for a format check to be used on the dates as given in the example data.

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[3]

Q12 - Page 8 0417/11 May/June 2016

8

Name and describe **three** methods of implementing a new computer system. For each one describe the type of situation where each method might be used.

1

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2

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3

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[9]

Q11 - Page 7 0417/12 May/June 2016

9

Different types of test data are used to test a newly developed system.

Complete the following sentences.

(a) Data which is of the correct type is called

..... [1]

(b) Data which is outside a given range is called

..... [1]

(c) Data which is at the boundaries of a given range is called

..... [1]

(d) Data which has been used with the current system and the results are known is called

..... [1]

Q4 - Page 3 0417/13 May/June 2016

10

Enrique has developed a new database system for a supermarket. Below is one record from the database. He needs to test the validation checks that he has developed for the system.

Barcode	Supplier_code	Contents	Price	Reorder_level	Weight(kg)
5012427141308	EZ123	Lobster Bisque	£2.72	100	0.4

- (a) He designed a length check for the **Barcode** field. However, when he entered the code 5012472141308 by mistake, the system still accepted it.

Describe the check he could have used to prevent this error.

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.....

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..... [2]

- (b) The **Reorder_level** for each item is never more than 100. He developed a validation check to make sure this is the case.

However, when he entered the number 100 it was rejected.

Name and describe the validation check he should use and explain why the number could have been rejected.

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..... [3]

- (c) He also designed a length check for the **Supplier_code** field. However, when he entered EZQ23 by mistake, the system still accepted it.

Describe how he could have prevented this error.

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..... [2]

- (d) For each field write down the **most appropriate** data type. For any numeric field specify the type of number.

Barcode.....

.....

Supplier_code.....

.....

Contents.....

.....

Price.....

.....

Reorder_level.....

.....

Weight(kg).....

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[6]

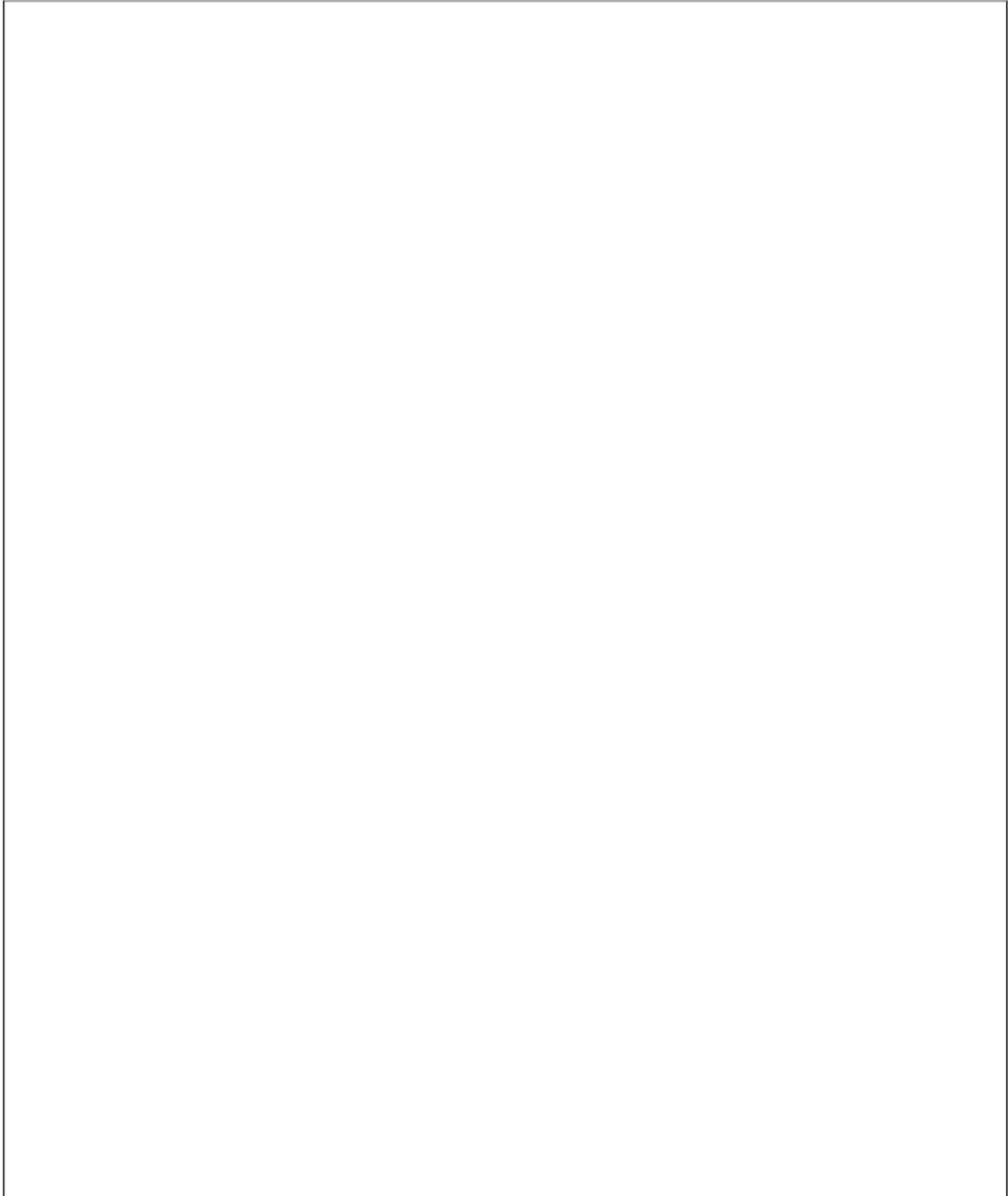
Q11 - Page 8 0417/13 May/June 2016

11	A new system has to be tested before it can be implemented. A variety of test data is used. Name the following types of test data. (a) Data that is outside the boundaries of acceptability. [1] (b) Data that is on the boundary of acceptability. [1] (c) Data that is not on the boundary but is within the range of acceptability. [1] (d) Data that has been used before and has known outcomes. [1] Q4 - Page 3 0417/11 May/June 2017
12	Validation is a way of checking data in a database. Name the most appropriate validation check that matches each description below. Makes sure that the data entered into a field is exactly 10 characters..... Makes sure that a number lies between 10 and 100..... Makes sure that the data entered is numeric..... Makes sure that the date is entered as DD/MM/YYYY..... [4] Q6 - Page 4 0417/11 May/June 2017

13

The Staywell Community Centre has created a new database which stores the details of each of its members. Each member can choose either swimming or tennis as an activity. The fields in the database are name, address, contact number, gender and activity.

- (a) When new members arrive at the centre they need to complete a paper-based form. Design a suitable paper-based form to collect the member's details and the activity they wish to book for.



[5]

(b) Describe **four** changes which could be made to this paper-based form to make it into an online screen form.

- 1.....
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- 2.....
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- 3.....
.....
- 4.....
.....

[4]

Q12 - Page 11 0417/11 May/June 2017

14

Tick whether the following items are found in **technical** documentation or **user** documentation or **both**.

	technical (✓)	user (✓)	both (✓)
Error messages			
Hardware requirements			
Program listing			
Purpose of the system			

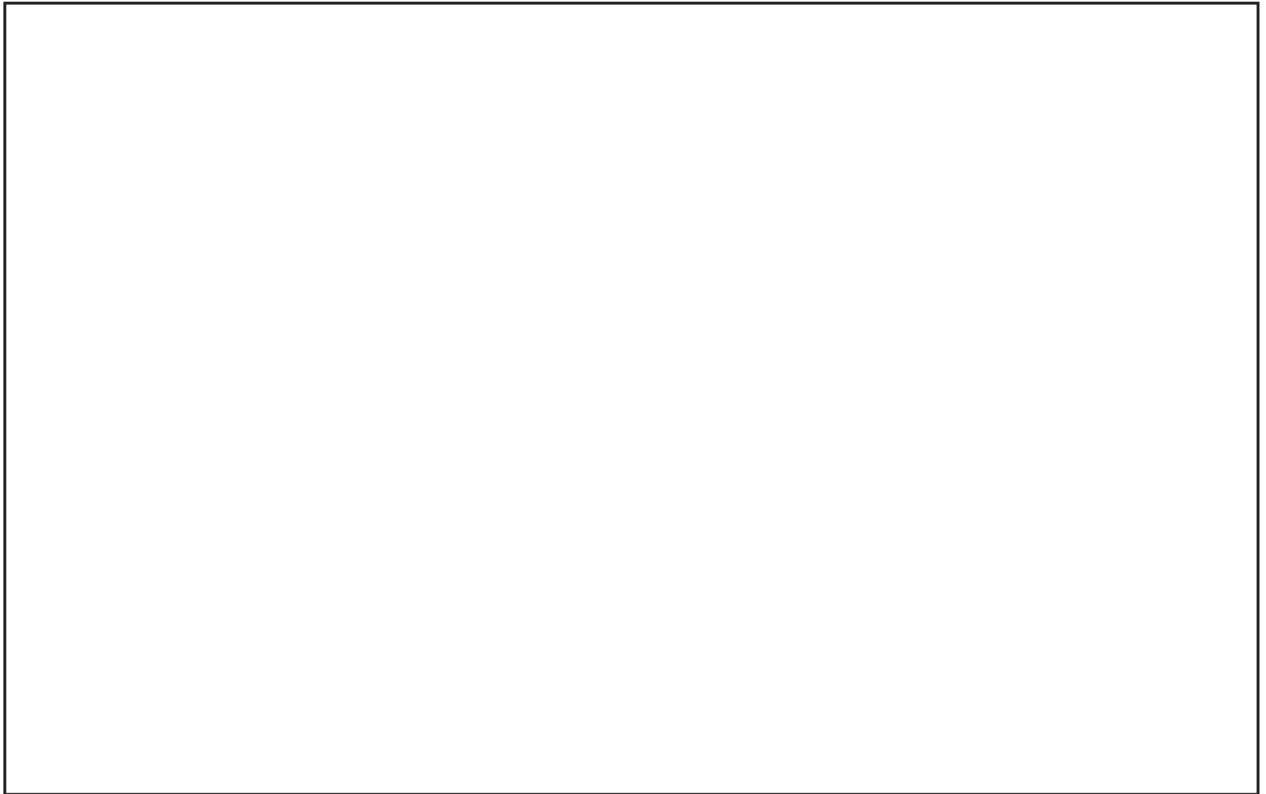
[4]

Q3 - Page 3 0417/12 May/June 2017

15

A leisure centre is creating membership cards for all its members. The membership card must include a photograph of the member, the member's name and membership number, the type of member (Senior or Junior), their date of birth and gender.

(a) Use the space below to design the membership card.



[5]

(b) Describe all the steps involved in carrying out a mail merge to create the membership cards.

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.....

[5]

Q10 - Page 9 0417/12 May/June 2017

16

When a new system is being developed analysis takes place. One method of researching the current system is interviewing.

(a) List **three** other methods of researching the current system.

- 1.....
- 2.....
- 3.....

[3]

(b) Describe **two** drawbacks of using interviews.

-
-
-
-
-
-
-
-

[4]

Q12 - Page 11 0417/12 May/June 2017

17

Systems can be implemented in different ways; one of these is direct changeover.

Name **three** other methods of implementation. For each method give an advantage when compared with direct changeover.

1

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2

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3

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[6]

Q11 - Page 11 0417/13 May/June 2017

18

A large organisation is introducing a new computer system.

Compare and contrast pilot implementation with phased implementation of the new system.

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[6]

Q8 - Page 6 0417/11 May/June 2018

19

Carlos is designing a new computer system to replace an existing system.

(a) Tick **four** items which will need to be designed.

	Tick (✓)
Inputs to the current system.	
Data capture forms.	
Report layouts.	
Limitations of the system.	
Observation methods.	
Improvements to the system.	
User and information requirements.	
Validation routines.	
Problems with the current system.	
File structure.	

[4]

(b) Before the system is implemented it needs to be tested. Different types of test data are used to test the system. An example of test data is live data.

Describe what is meant by live data.

.....

.....

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.....

[2]

(c) Following the implementation of the system, technical documentation needs to be written.

Identify **three** components of technical documentation which are not found in user documentation.

1

2

3

[3]

Q10 - Page 9 0417/11 May/June 2018

20

A new electronic cricket scoreboard is being tested. It stores the number of runs scored by each player during a cricket match. The designer of the scoreboard assumes that no player will score more than 400 runs.

Describe each type of test data shown below giving an appropriate example.

(a) Normal

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..... [2]

(b) Abnormal

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..... [2]

(c) Extreme

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..... [2]

Q15 - Page 13 0417/11 May/June 2018

21	When the new system has been developed, both user and technical documentation have to be written. Name three components that are found in both technical and user documentation. 1 2 3 [3]
Q10 - Page 11 – C 0417/12 May/June 2018	
22	When a system has been developed, two types of documentation are produced, technical and user documentation. (a) Explain why user documentation is needed. [2] (b) Explain why technical documentation is needed. [2] (c) Identify four components that are only found in user documentation but not found in technical documentation. 1 2 3 4 [4]

- (d) Tick three evaluation strategies that need to be carried out following implementation of the new system.

	Tick (✓)
Observe users operating the old system.	
Compare the final solution with the original requirements.	
Design the report layout.	
Check user documentation to see if it is correct.	
Interview users to gather responses about how well the new system works.	
Test the system works correctly.	
Identify any necessary improvements that need to be made.	
Design error handling.	

[3]

Q11 - Page 10 0417/13 May/June 2018

23

A small company makes toys and then delivers them to shops.

Throughout the day orders are received by the company from its customers. The office workers in the finance department create and store an invoice for each order. They are too busy to be disturbed from their work.

Delivery drivers receive copies of the invoices which they will pass on to their customers. The drivers make a large number of deliveries per day and do not return to the office.

A systems analyst will research the current system and suggest improvements to be made.

- (a) For each type of employee identified above, describe the most suitable method of collecting information from them, giving a reason for your choice.

Office workers.....

.....

Reason.....

.....

Delivery drivers.....

.....

Reason.....

.....

[4]

- (b) The factory is based on one site and only has space for one production line. The company has decided to computerise its production line. It will use direct changeover as the method of implementing the new system.

Explain why the company would use this method rather than other methods of implementation.

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[4]

Q17 - Page 14 0417/13 May/June 2018

24

The systems life cycle is followed when a new computer system is being created.

Tick whether the following statements are examples of the **Analysis** stage, the **Design** stage or the **Evaluation** stage of the systems life cycle.

	Analysis (✓)	Design (✓)	Evaluation (✓)
Identifying the problems with the current system			
Deciding on testing strategies			
Observation of workers using the current system			
Comparing the solution with the original task requirements			

[4]

Q10 - Page 12 0417/12 May/June 2019

25

Tawara Health Centre is creating a new online appointment system for its patients.

The developer has created a table showing part of the file structure of the new database.

Field name	Data type	Field length	Comment
Medical_ID_number	Numeric: integer		12-digit unique number
Date_of_birth	Date/Time		dd/mmm/yyyy
Family_name	Text	10	
Doctor_name	Text	20	
Appointment_time	Date/Time	5	hh:mm
Contact_telephone_number	Text		01617151234
Current_medication	Text	20	
Blood_type	Text	3	A, O, AB, B Each can be + or –

- (a) Apart from presence check, for each field, name an appropriate validation check that could be used. Your answer should be different in each case.

Field name	Validation check
Medical_ID_number	
Blood_type	
Date_of_birth	
Contact_telephone_number	

[4]

- (b) Referring to the fields in the table, name the most appropriate field that could be used as a primary key.

..... [1]

- (c) Using the fields given in the database complete the design below showing a suitable online data entry form for the receptionist to fill in.

TH

[6]

Q8 - Page 6 0417/13 May/June 2019

26

A hospital is creating a database to store blood donors' medical details. The fields in the database include Health_number, Date_of_birth, Gender and Blood_group.

(a) When a new blood donor wishes to give blood they need to fill in a form.

Each blood donor has a unique health number which uses a 10-digit number, for example 9434765919.

Blood group can be A, B, AB or O.

Part of the form is shown below.

Blood donor

1. What is your health number?

2. What is your date of birth? / /

3. What is your gender? M/F

4. What is your blood group?

Name a suitable data type for the answers given to each of the questions in the form. For any numeric field, specify the type of number.

Question 1.....

Question 2.....

Question 3.....

Question 4.....

[4]

(b) In the Blood_group field the only data that can be entered is A, B, AB or O.

Tick (✓) whether each of the following blood groups is an example of **abnormal** or **normal** data.

Blood group	abnormal (✓)	normal (✓)
AB		
C		
B		
AO		

[2]

(c) The administrator needs a list of all the blood donors in the database with blood group A as well as in the same list all the blood donors with blood group AB. This list will only include female blood donors.

To find the blood donor with the health number 9434765919 he will need to type in the following query.

Health_number = 9434765919

Write a query to find all the female blood group donors with the blood group A as well as those with AB.

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[6]

(d) When the details of a new donor are added to the database the person's date of birth is entered.

Explain why the administrator would prefer their date of birth to be stored rather than their age.

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[4]

(e) Name the most appropriate validation check for the Health_number.

.....

.....

[1]

Q4 - Page 4,5,6 0417/11 May/June 2020

The network manager at a school has been asked to develop a student registration system to help staff keep a record of student attendance.

[illegible]

[6]

A teacher is creating a spreadsheet that will record the grades students achieved in recent tests and compare the average grades of these tests with the students' forecast grades. The grading the teacher is using awards 1 for the highest grade down to 10 for the lowest. The teacher has created a validation rule on the grade entered.

Type of test data	Example of test data
Normal	
Abnormal	
Abnormal	
Extreme	

[4]

Q4 – Page5 0417/12 May/June 2020

29

The secretary of the Tawara Rowing Club is organising a presentation evening for its members. A systems analyst is creating a database for the club to store the details of the members. She is also creating a spreadsheet to show who is attending the presentation evening.

The systems analyst has set up the following fields for the database.

Name_of_person	Date_of_birth	Membership_type	Contact_email	Contact_phone_no	Year_joined	Gender
Nor Kwa	2/4/2005	Social	n.kwa@rockict.com	03 2453 5673	2018	F
Adam Mazian	23/5/2003	Social	AdamM@abc.co.my	082 25 4689	2016	M
Ahmed Othman	12/03/2006	Junior	AOthman@cie.org	01223 432678	2018	M
Zara Png	1/12/1997	Senior	Zara@cbc.cn	123 3267 9999	2010	F

- (a) For the following fields write down the most appropriate data type. Each data type must be different. For any numeric field, specify the type of number.

Gender.....

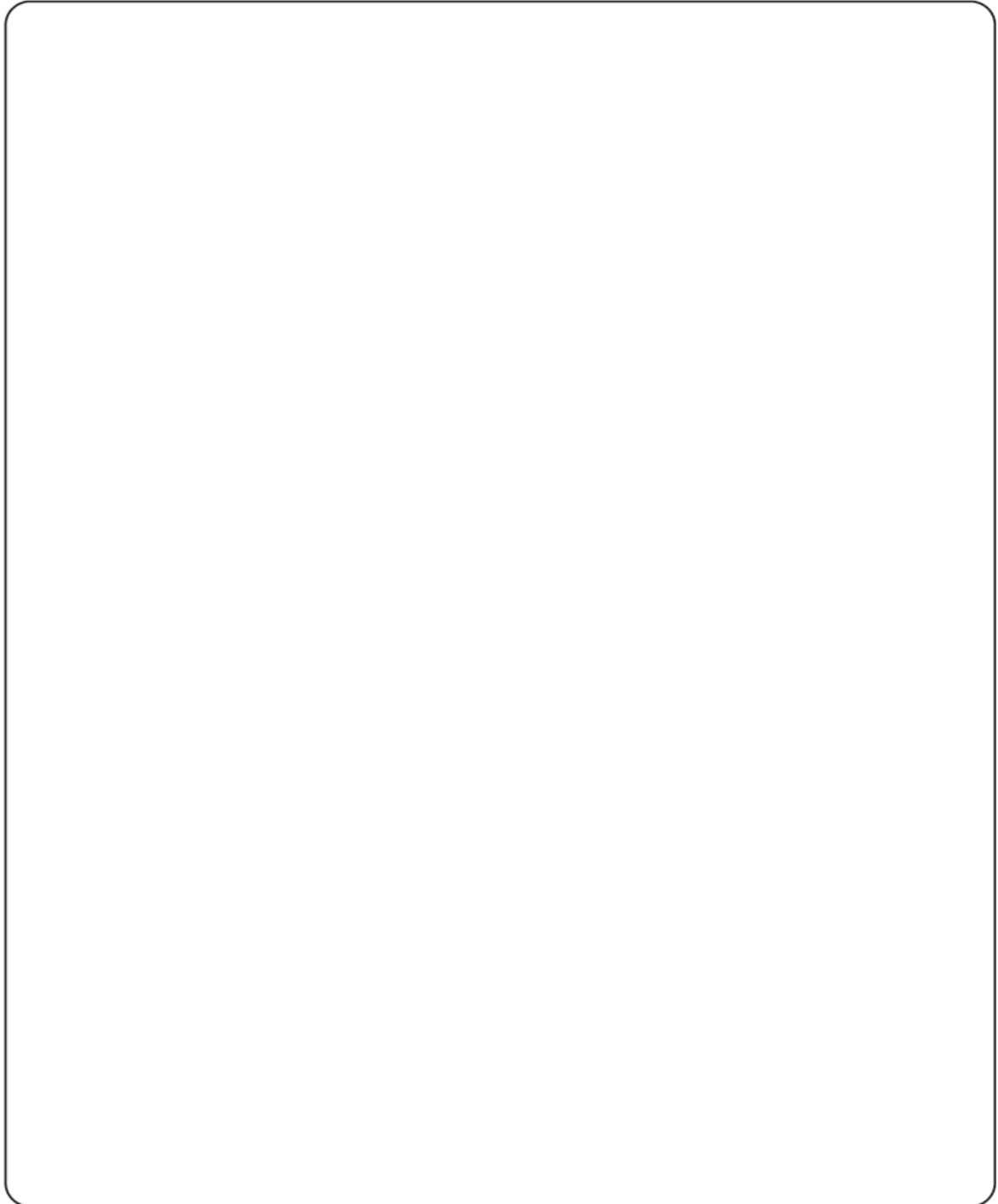
Membership_type.....

Year_joined.....

[3]

- (b) Design an online data entry form which could be used to enter the details of one member, using all the fields shown in the table.

In your design include appropriate spacing for each field and navigational aids. Do not include data in the form.



[8]

Q4 – Page4 a,b 0417/13 May/June 2020

30

Parallel running is a type of implementation.

Name and describe **two** other methods of implementing a new computer system. For each one give an advantage and a disadvantage of this method. Your answers must be different in each case.

Method 1.....

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Advantage 1.....

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Disadvantage 1.....

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Method 2.....

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.....

.....

Advantage 2.....

.....

Disadvantage 2.....

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[8]

Q11 – Page15 0417/13 May/June 2020